ShopEase User Management Service

FastAPI User Profile and Address Microservice with JWT Validation, MySQL and Docker

Submitted By: Dineshkumar S

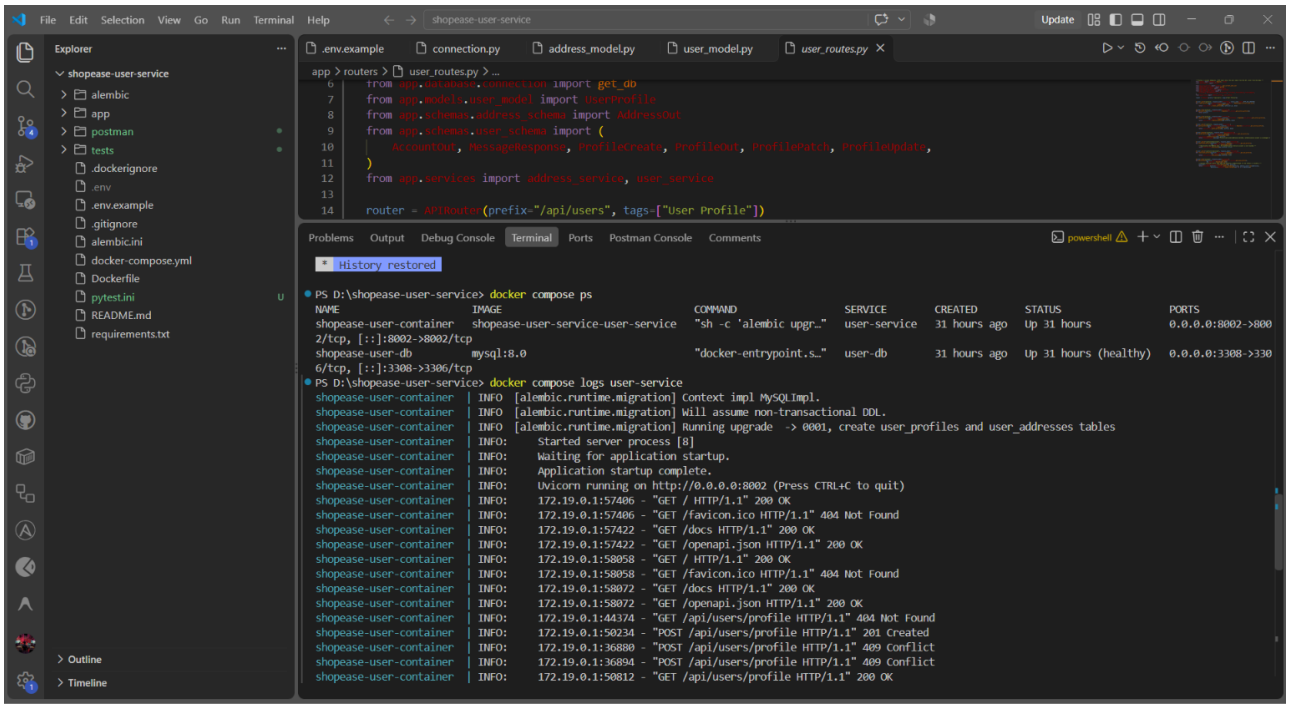
Date: 11/11/2026

Requirement Completion

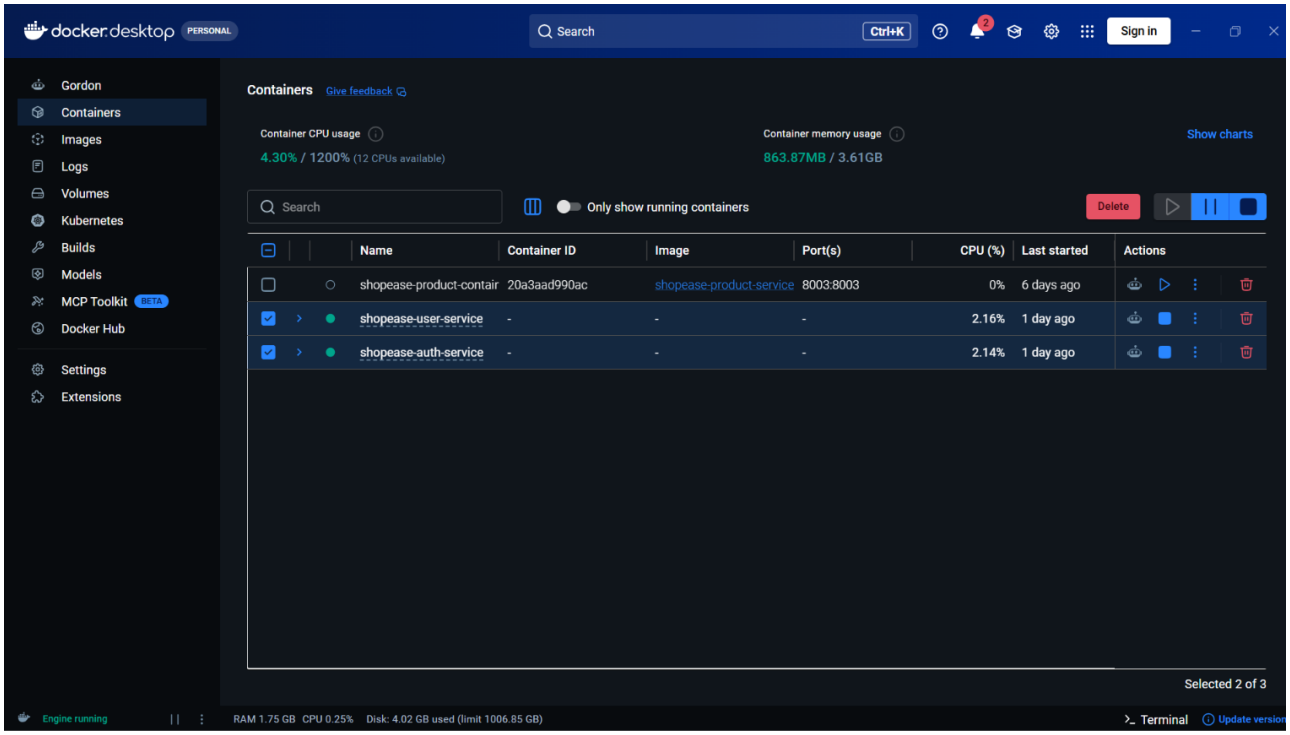
Requirement	Status	Evidence
User Service Source Code	**Done**	FastAPI project with separate folders for routers, services, models, schemas and database. Runs on port 8002.
MySQL Database	**Done**	Separate MySQL 8 database shopease_user_db in its own container (host port 3308), with user_profiles and user_addresses tables and a unique auth_user_id.
Alembic Migration	**Done**	Migration 0001 creates the user_profiles and user_addresses tables and runs automatically when the container starts.
JWT Authentication	**Done**	Every endpoint needs a Bearer token issued by the Auth Service. Missing token returns 401 “Bearer token missing” and an invalid token returns 401 “Invalid token”.
Profile APIs	**Done**	Create (201), duplicate rejected (409), get, update and a missing profile returns 404. The user ID comes from the token, not the request body.
Address APIs	**Done**	Add (201), update (200) and delete (200) addresses, all linked to the profile owner.
Separate User Profiles	**Done**	User B logs in with a separate token and gets its own profile (auth_user_id 5, id 2), separate from User A (auth_user_id 4, id 1).
Dockerfile and Docker Compose	**Done**	Runs user-service and user-db (mysql:8.0) with a MySQL health check; both containers are up and the database is healthy.
Environment Variables and Secrets	**Done**	Database credentials and JWT settings are in .env, which is excluded from Git and the Docker build.
API Testing	**Done**	Scenarios 1 to 6, 8, 10, 13 and 14 tested in Swagger UI with the expected status codes. Requests also run in Bruno from the Postman collection.
README Documentation	**Done**	Overview, setup, Alembic and Docker commands, endpoints and test results.
GitHub Repository	**Done**	Source code pushed to https://github.com/Dineshkumarsenthil/shopease-user-service

Evidence

Docker: Containers Running (docker compose ps) and Application Logs



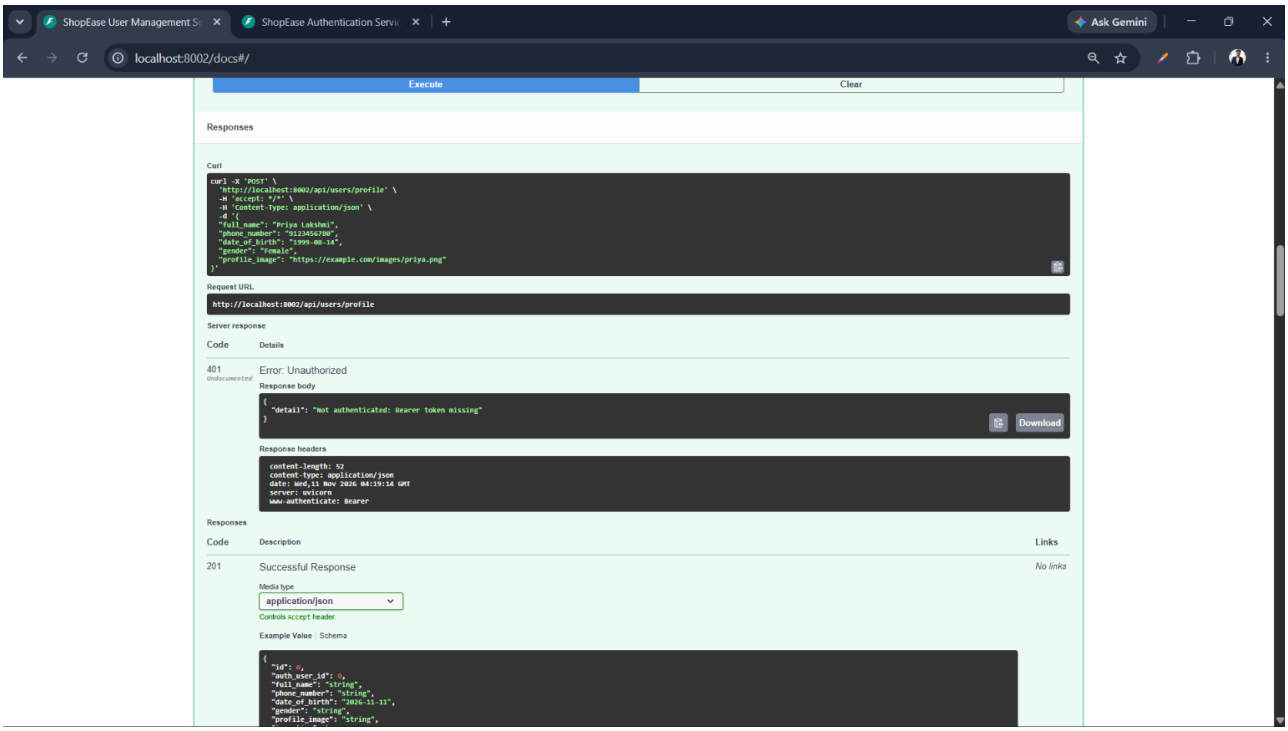
Docker Desktop: User Service and Auth Service Containers Running



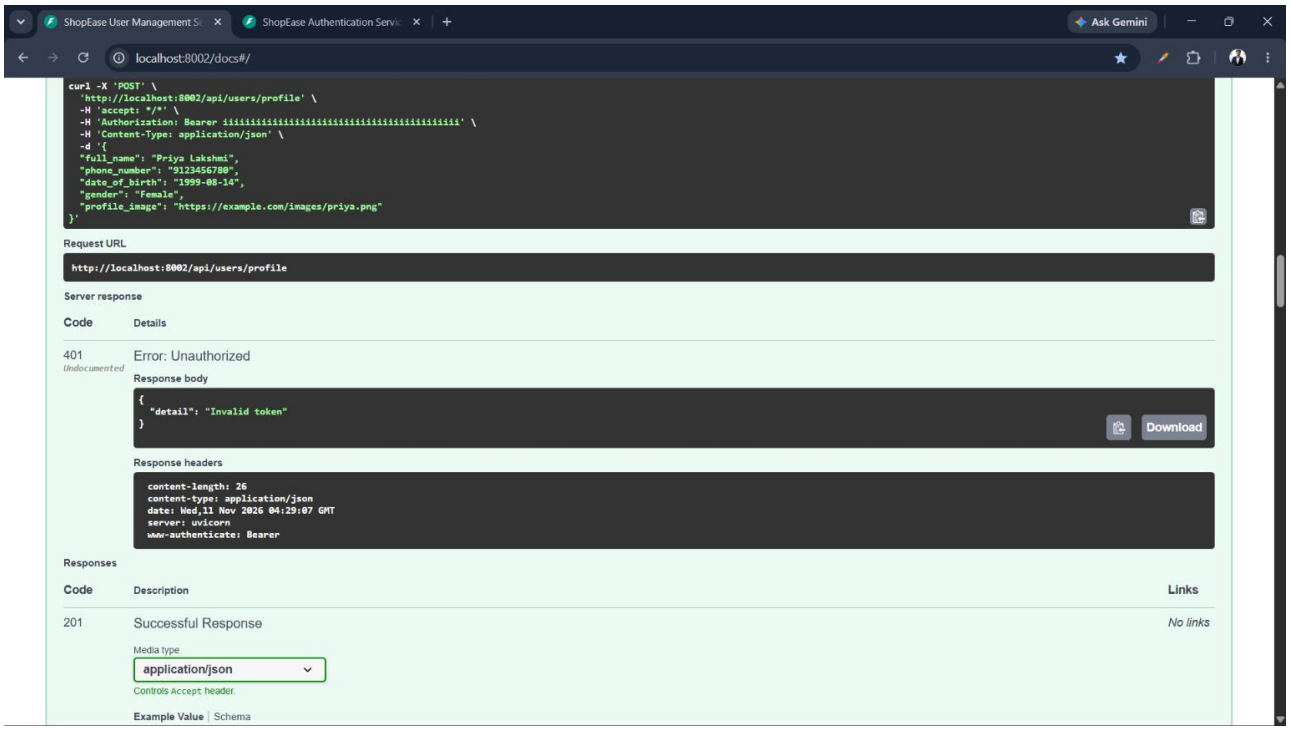
API Documentation: Swagger UI Endpoint List



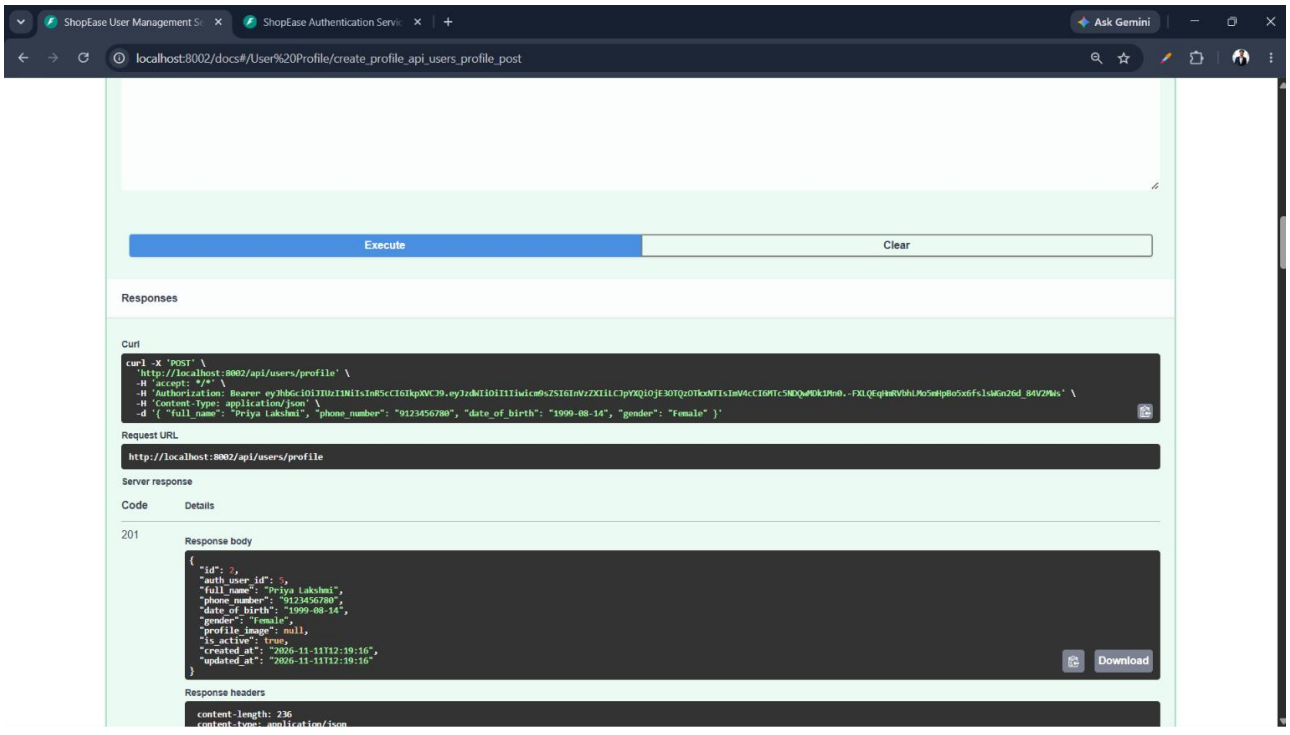
Scenario 1: Request without Token, 401 Unauthorized



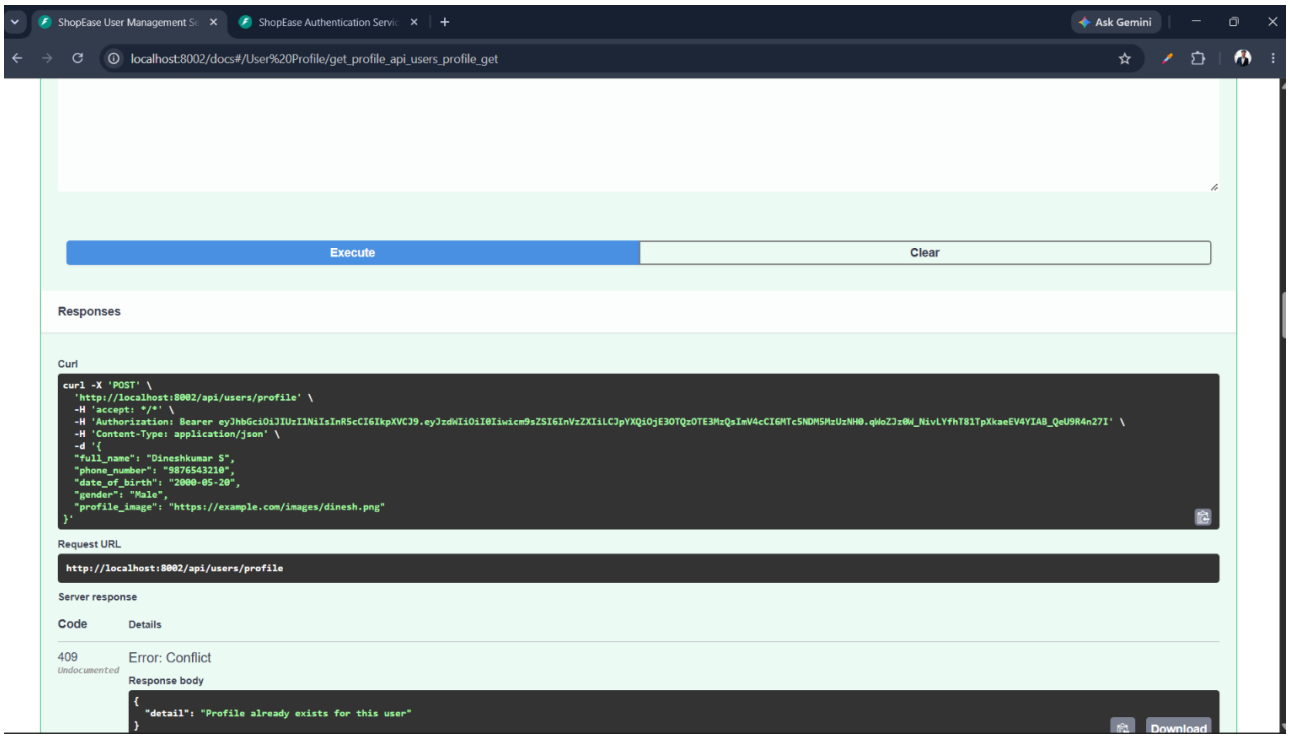
Scenario 2: Request with Invalid Token, 401 Unauthorized



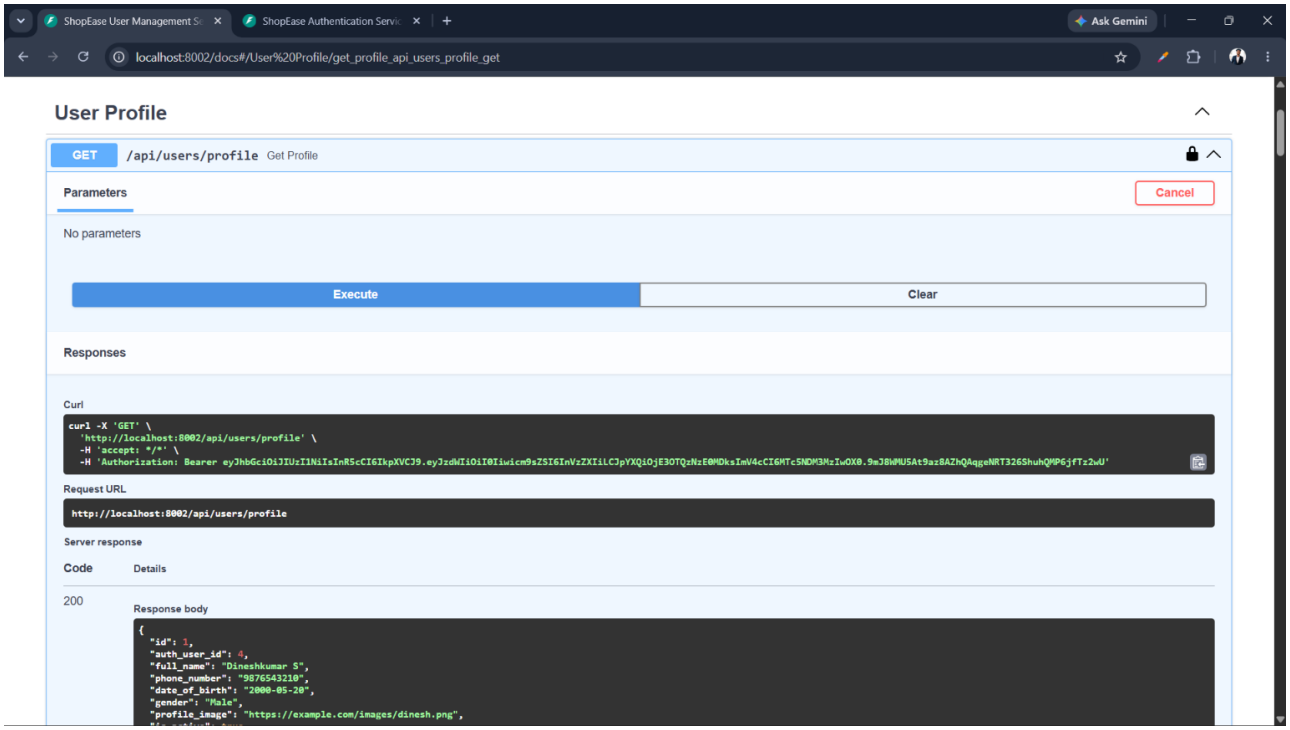
Scenario 3: Create Profile, POST /api/users/profile, 201 Created



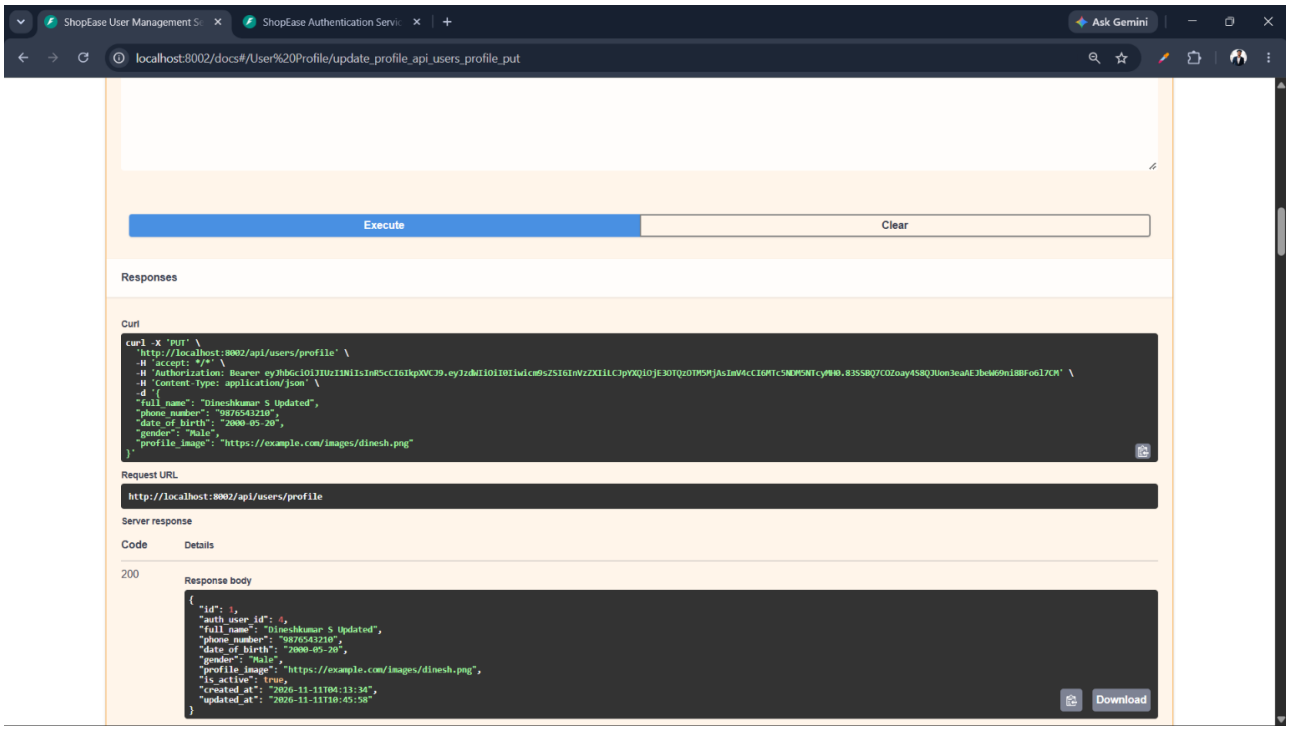
Scenario 4: Duplicate Profile, 409 Conflict



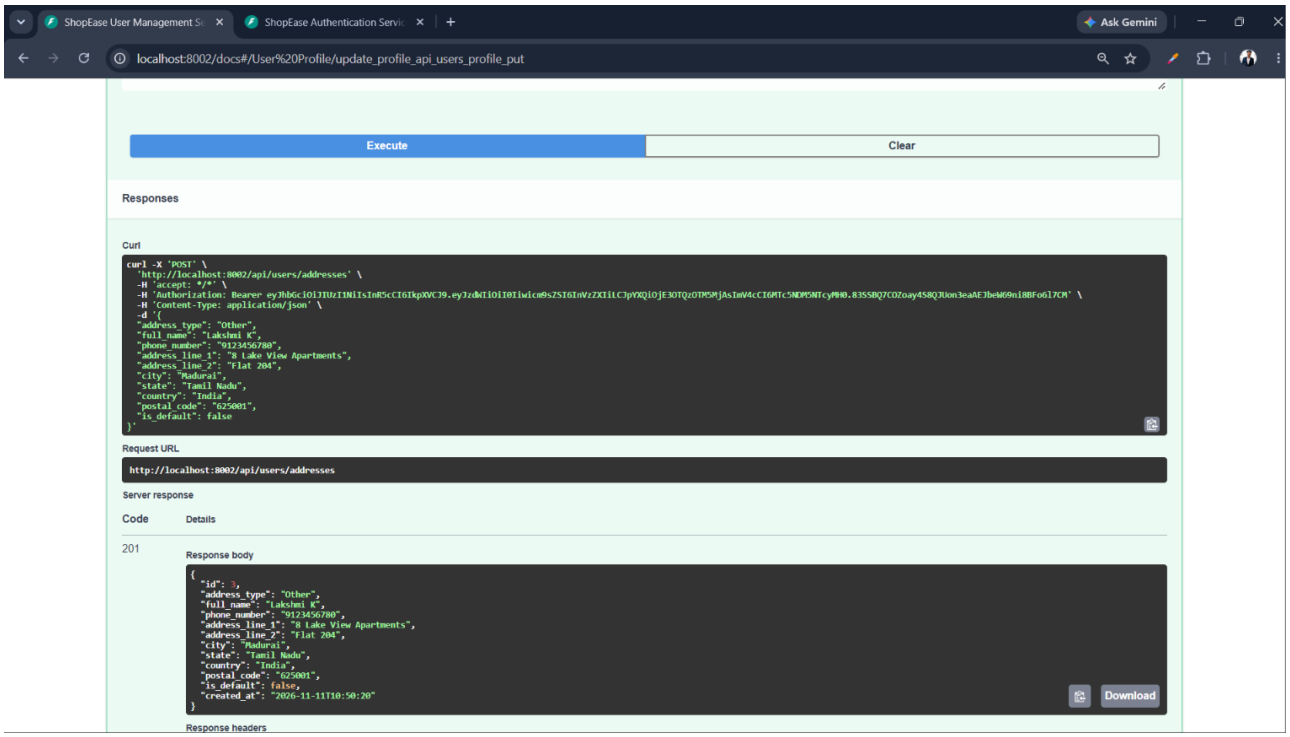
Scenario 5: Retrieve Own Profile, GET /api/users/profile, 200 OK



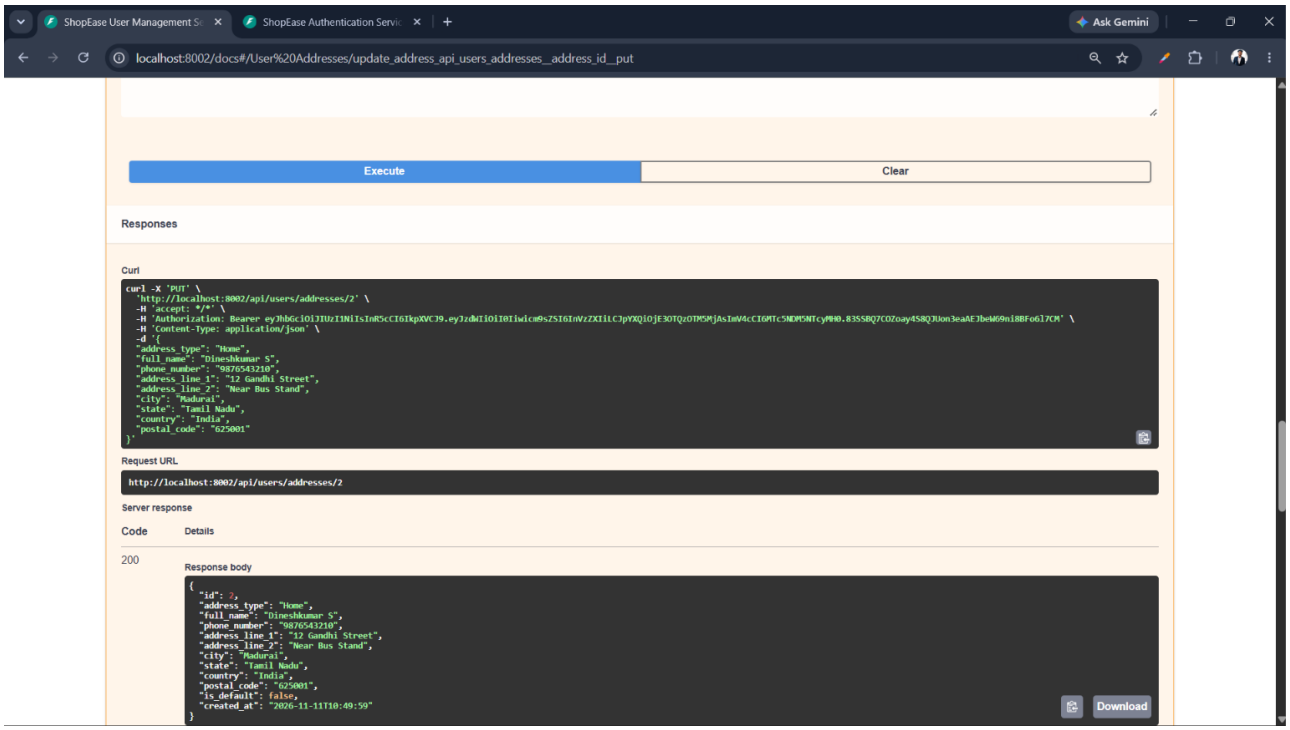
Scenario 6: Update Profile, PUT /api/users/profile, 200 OK



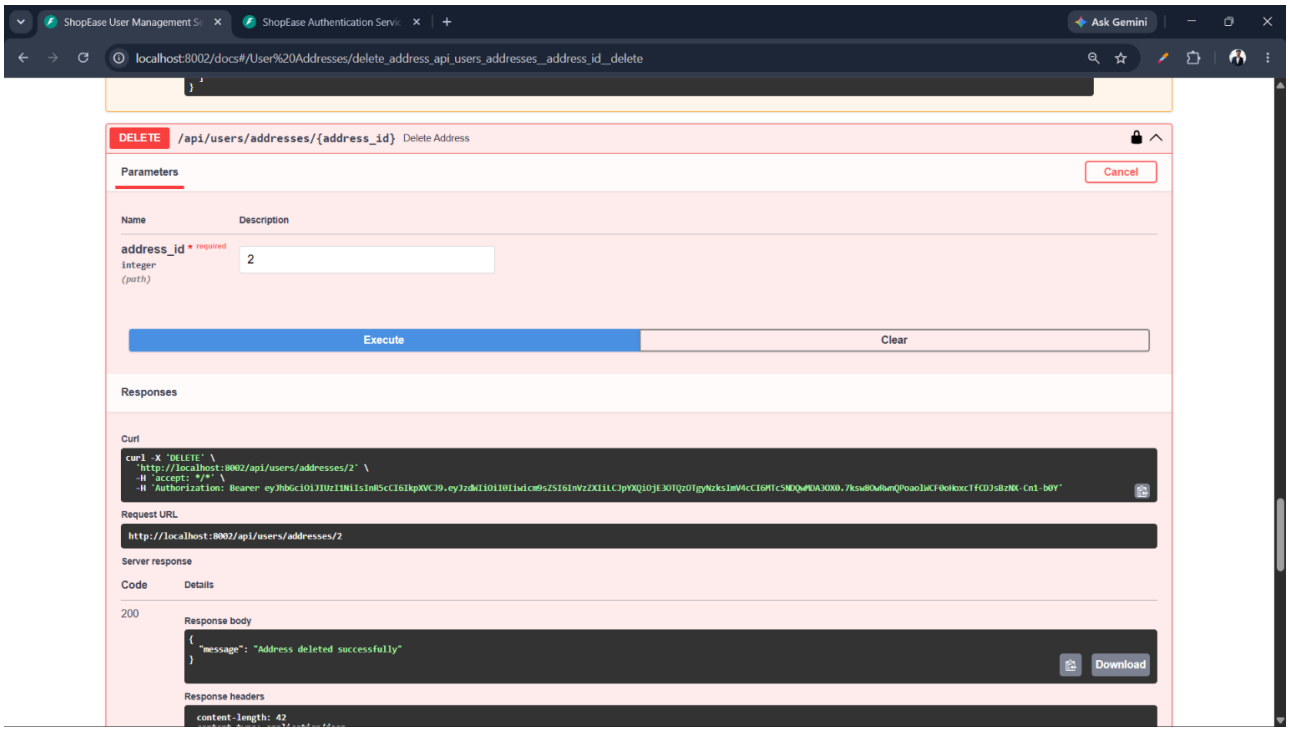
Scenario 8: Add Address, POST /api/users/addresses, 201 Created



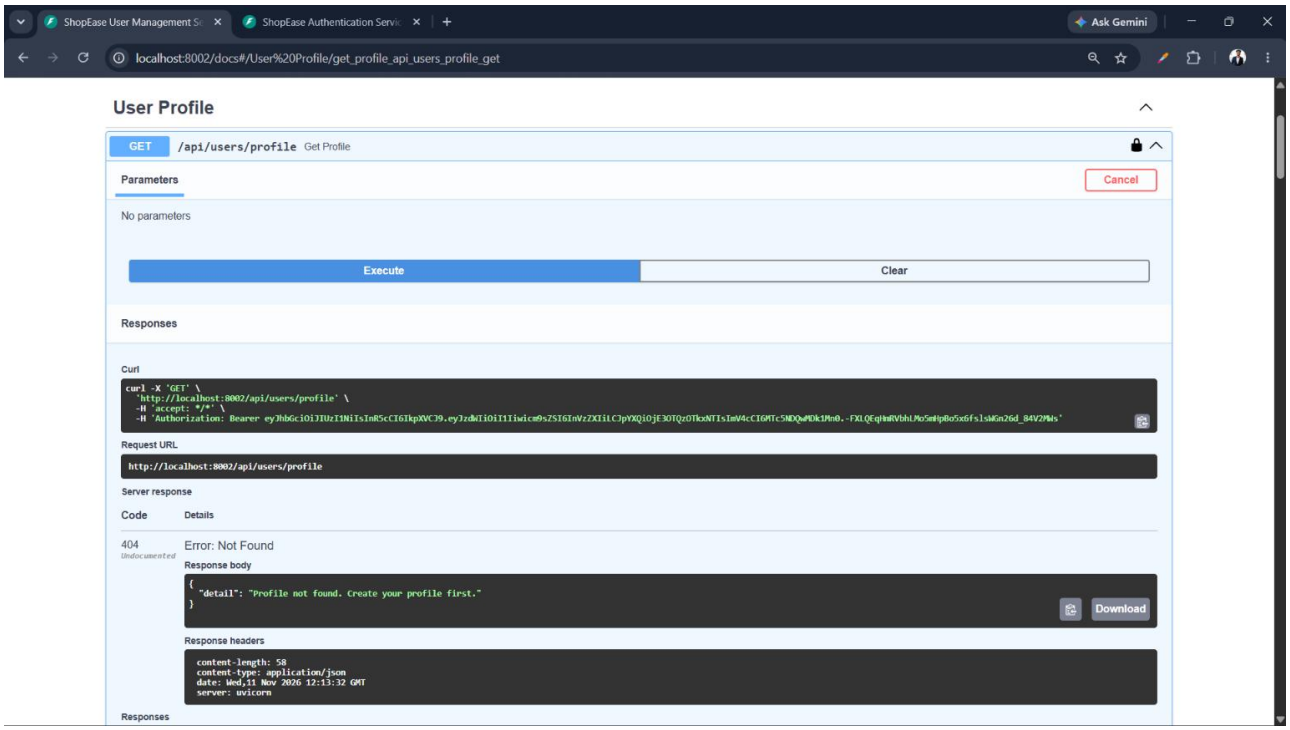
Scenario 10: Update Address, PUT /api/users/addresses/2, 200 OK



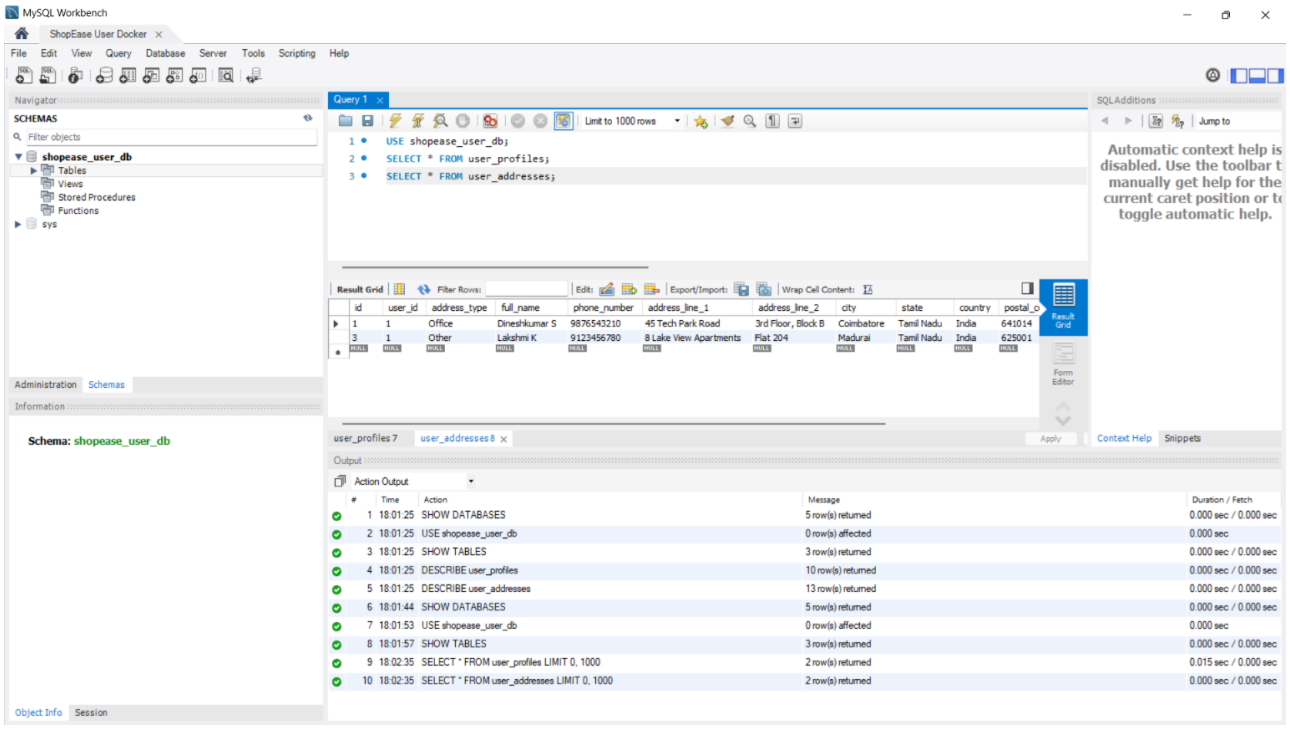
Scenario 13: Delete Address, DELETE /api/users/addresses/2, 200 OK



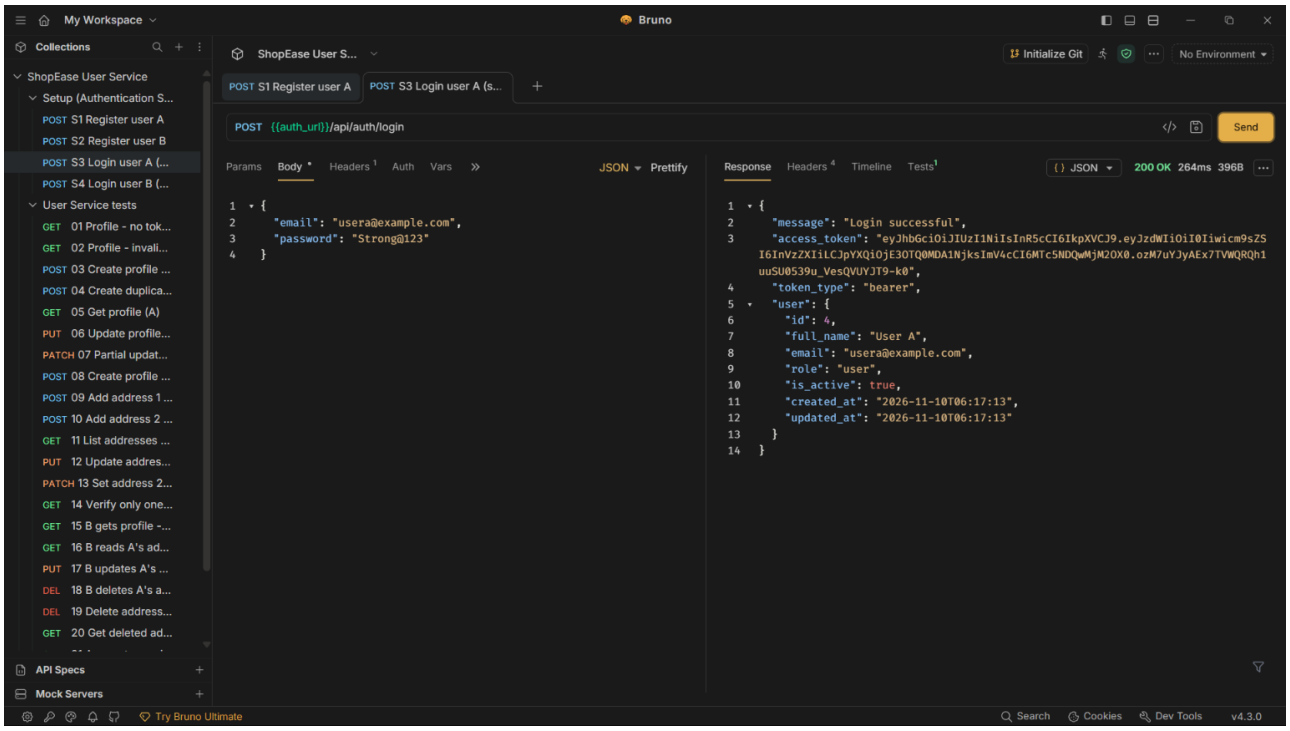
Scenario 14: User B Cannot Read User A's Profile, 404 Not Found



Database: user_profiles and user_addresses in shopease_user_db



API Client Testing: Login Request Run from the Imported Collection (Bruno)



API Client Testing: Get Profile Request Run from the Imported Collection (Bruno)

